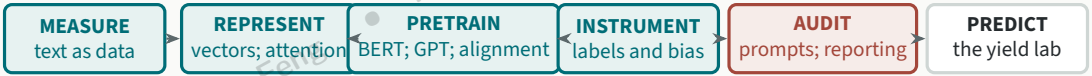
Session 5.5

Simulated Agents, Prompts as Design, and Reproducible LLM Research

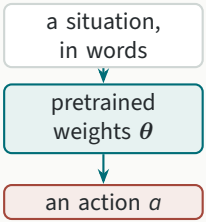
What a model can stand in for, why the prompt is a confounder, and what has to be reported



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Homo Silicus: A Subject You Can Query



no parameter is estimated

behaviour is read out

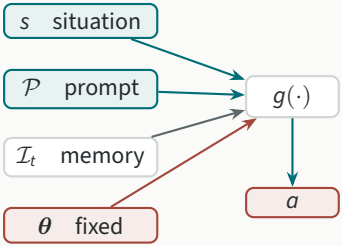
the weights hold the behaviour text *records*

- **Homo silicus**: the model as an implicit computational model of human behaviour, queried in prose rather than estimated.
- It took fairness, reciprocity and loss aversion from human writing, so it answers a classic game in the human direction.
- **But nothing separates generalisation from recall of the published experiments.**

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Source: Horton (2023), homo silicus; Miao & Feng, *AI for Economists*, ch. 14, “LLM 作为模拟经济主体”.

The Agent's Policy: What You Move, and What You Cannot



two of the four are prose you write

- Behaviour is a conditional distribution over actions:

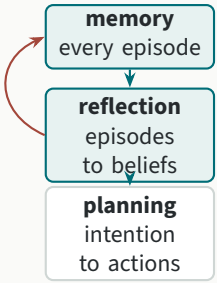
$$a \sim g(\cdot \mid s, \mathcal{P}, \mathcal{I}_t; \boldsymbol{\theta}).$$

- You move two pieces of prose, s and $\mathcal{P}$, not the parameters of a utility function.
- $\boldsymbol{\theta}$ is fixed, so all behavioural heterogeneity enters through the role instruction.
- **The role instruction does the work a preference parameter used to do.**

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Source: Miao & Feng, *AI for Economists*, ch. 14, "生成式智能体模拟".

Memory, Reflection, Planning: Where $\mathcal{I}_t$ Comes From

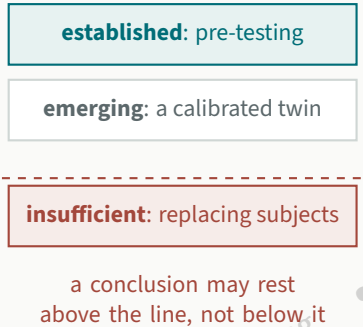


- The base model is stateless. Whatever must persist across rounds lives in $\mathcal{I}_t$, outside the model.
- Retrieval scores memories on importance, relevance and recency; reflection turns episodes into beliefs.
- **How many memories, at what weights, how often reflection runs: design parameters. Undisclosed, two runs are not comparable.**

Source: Park, O'Brien, Cai, Morris, Liang & Bernstein (2023), generative agents.

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Three Tiers of Evidence, and What Each One Licenses

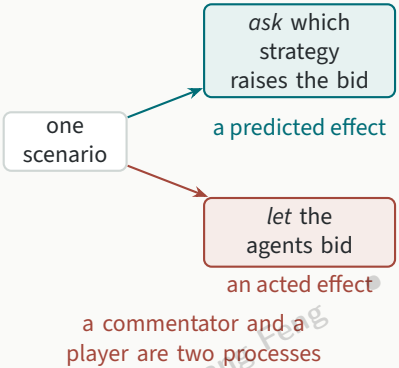


- **Established**: screening treatment variants. It touches the design, never the inference.
- **Emerging**: a twin calibrated on a long interview, reproducing one person's answers on particular scales.
- **Insufficient**: silicon subjects as the main source for a causal claim.
- **The line is conclusion strength, not model power: the same simulation is tier one screening and tier three estimating.**

Source: Miao & Feng, *AI for Economists*, ch. 14, "按证据成熟度划分应用场景".

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Three Threats to External Validity, and a Fourth Difference



- **Memorised replication**: human-like play in the classic games may be recall of published results. Test on a design the corpus never saw.
- **Corpus representativeness**: English and Western text dominate, and the substitution leaves no trace in the output.
- **Version instability**: "we used model X" is an incomplete method description.
- **A predicted effect is not an acted one: the unit of analysis is behaviour.**

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Source: Manning, Zhu & Horton (2024), automated social science; Argyle et al., *Political Analysis* (2023).

How to Write the Sentence, and Whose Consent It Needs

"the model *has* a fairness preference"

a theoretical claim: no one can check it

"under prompt $\mathcal{P}$ it produced outputs rejecting low offers"

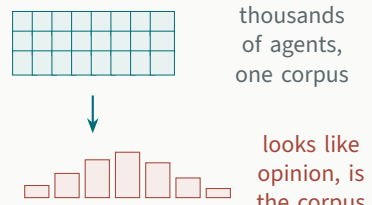
an observable: the same prompt reproduces it

- Talking to it in prose is **functional anthropomorphism** and is fine. Claiming it *has* an internal state is an unsupported assertion.
- **Consent**: a twin built from one interview extends that person's own data, so an identifiable simulation needs consent even with no human in the room.
- **"Analysis for this study" does not cover building a twin and querying it for years.**

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Source: Miao & Feng, *AI for Economists*, ch. 14, "伦理考量".

Synthetic Consensus, and What a Pre-registration Would Hold



- **Synthetic consensus**: a large agent network's output can look like a distribution of opinion while being the corpus's content preferences.
- Keep a human check at every policy-relevant node, and verify against the population.
- **Pre-registration, unsettled**: is temperature the randomisation? Is the sample runs, agents or scenario pairs? Does a causal model written in advance replace a pre-analysis plan?

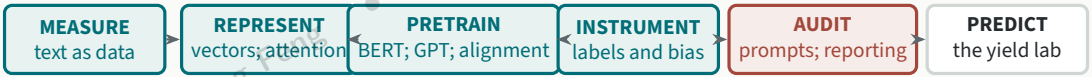
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Source: Miao & Feng, *AI for Economists*, ch. 14, "合成共识".

Session 5.5

The Model in Your Own Process

Where it changes the cost of the work, and where it must not touch the judgement



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Five Task Modules, Ordered by What You Can Hand Over

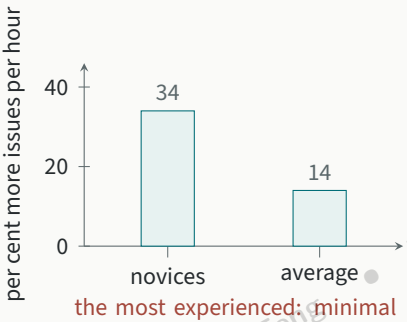
module	what it covers	how far you can lean on it
literature	retrieval and comparison frames	Far: a wrong citation shows up on checking
data	formats, matching, cleaning	Far, with assertions: nothing raises
code	prototypes, translation, errors	Far: running it exposes most errors
writing	drafts, rewriting, captions	Far: the author owns every word
identification	question, strategy, causation	**Not at all**: institutional knowledge

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- The first four are repetitive cognitive labour; the fifth is the academic contribution.
- **The red line: once a step's output decides the identification strategy, that step is yours.**

Source: Miao & Feng, *AI for Economists*, ch. 14, table of research tasks; Korinek, *JEL* (2023).

What the Productivity Evidence Actually Shows

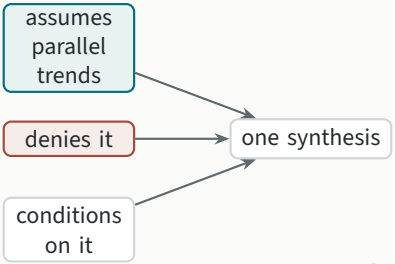


- Writing tasks, 453 professionals randomly given access: time down 40 per cent, quality up 18 per cent, the spread narrower.
- A standardised server-programming task: 55.8 per cent faster.
- Customer support, 5,179 agents: 14 per cent more issues per hour, 34 per cent for novices.
- **Cognitive levelling: the gain sits at the inexperienced end.**

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Source: Noy & Zhang, *Science* 381(6654):187–192 (2023); Peng et al. (2023); Brynjolfsson, Li & Raymond, *QJE* (2025).

The Synthesis Trap

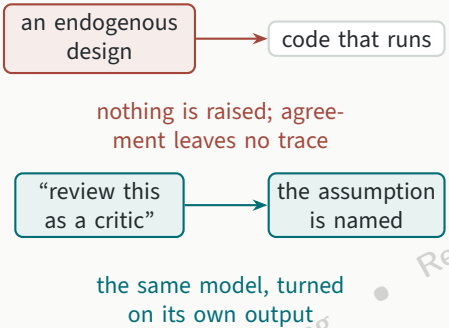


- Past a few hundred papers the bottleneck moves from finding to integrating.
- The model smooths theoretical disagreement into an over-harmonious narrative. Alignment prefers uncontested statements, so a better prompt does not remove it.
- **Invert the task**: ask which papers disagree on the same identifying assumption and who is on each side.

Source: Miao & Feng, *AI for Economists*, ch. 14, "整合的陷阱".

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It Will Go Along with a Flawed Design



- **Silent errors**: a merge that drops rows on a case mismatch, a conversion the wrong way round, a window off by one. They propagate as bias.
- **Leakage**: pretraining holds the outcome, so the sentiment of a past date comes back contaminated by hindsight.
- Hand it an endogenous design and it writes working code rather than naming the problem.
- **Assertions after every step, and the model turned on its own output.**

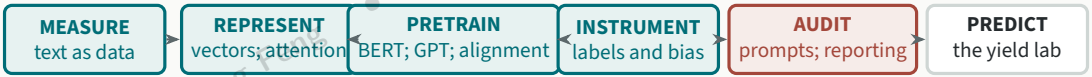
Source: Miao & Feng, *AI for Economists*, ch. 14, “数据与代码环节的两类静默风险”.

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Session 5.5

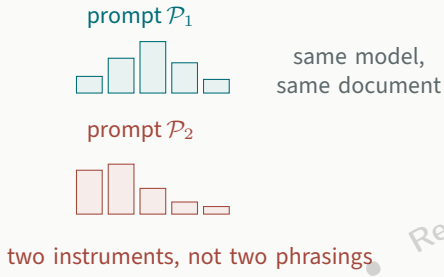
The Prompt Is the Instrument

Why the wording is a design variable, and what this course measured on its own



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The Prompt Is a Confounder, Not an Interface



- Hold the model and the input fixed. The prompt still moves the output distribution:

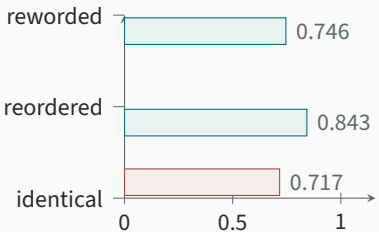
$$\partial\, \mathbb{E}[\hat{m}_i \mid \mathbf{x}_i,\, f_v] \,/\, \partial\, \mathcal{P} \;\neq\; 0.$$

- A criterion, not a model. Where that derivative is non-zero the prompt belongs in the methods section beside the sample rule.
- **The prompt does the job the instruction sheet does in a laboratory experiment.**

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Source: Miao & Feng, *AI for Economists*, ch. 14, "提示选择作为混淆变量".

This Course Measured Its Own Instrument



one model only, so agreement *across* models is not in this run

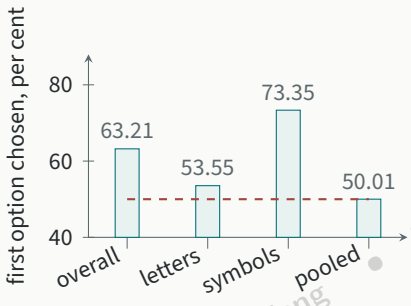
- One frozen prompt on 245 statements; three variants on a stratified 48: B reworded, C reordered, **D byte-identical to A**.
- Agreement with A: 83, 90 and 81 per cent. Not pinned: the harness's own system prompt, and the temperature.
- **The identical prompt agrees with itself least. Without D, B's gap reads as wording — and reporting the best prompt is *p*-hacking.**

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claude-sonnet-5 via Claude Code 2.1.278 · 2026-09-20 · prompt 846fe144

Source: This course's own recorded run: 389 headless calls on the frozen statement corpus, 2026-09-20.

Primacy: The Number, and the Direction

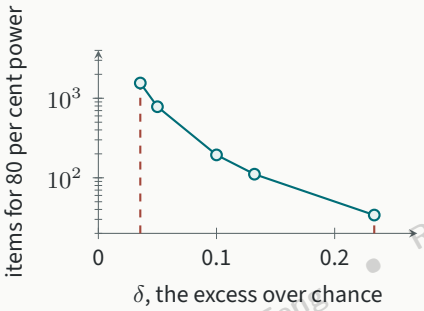


- On a task whose answer should be fifty-fifty, the first option is chosen 63.21 per cent of the time, and saying the order is random does not remove it.
- Pooling over the full factorial of 32 architectures returns 50.01 per cent: randomisation, not exhortation.
- **The direction surprises people**: *symbol* labels give 73.35 per cent against 53.55 for letters, and reviews often print it reversed.

Source: Brucks & Toubia, *PLOS ONE* 20(4):e0319159 (2025), full-factorial design.

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The Power Calculation That Settles the Robustness Check



- Let δ be how far the first-option rate sits above chance. Two-sided at 5 per cent, 80 per cent power:
- Letter labels, $\delta = 0.0355$: 1,555 items. Symbol labels, $\delta = 0.2335$: 34. A 46-fold difference from formatting alone.
- On 50 items the power against the smaller effect is 0.10.
- **So “no order effect on 50 items” is evidence of nothing.**

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Source: Miao & Feng, *AI for Economists*, ch. 14, design calculation calibrated on Brucks & Toubia (2025).

Three Design Principles

one boundary examples	the dividing line, not the centre
two reason, then answer	a trace you can audit
three call out, do not recall	external facts beat stale memory

- **Put the boundary in the examples**: demonstrate the hard cases near the dividing line, not the clearest positives.
- **Reason first, answer second**: a sentence of justification before the label lowers boundary errors and leaves a trace.
- **Call out rather than recall**: retrieval replaces the stale memory that fabricates.
- **The tool environment is part of $\mathcal{P}$, so disclose it with the prompt.**

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Source: Brown et al. (2020); Wei et al. (2022); Kojima et al. (2022); Yao et al. (2023).

Temperature, Drift, and Four Things to Report

- **Temperature has two opposite uses.** For annotation, zero: one input, one measure. For simulation, zero is harmful — every simulated subject answers identically.
- For simulation, treat sampling as Monte Carlo randomisation: draw R times and report the distribution. The book marks this as its own recommendation, not a settled finding.
- **Version drift**: a named service changes behaviour between releases and you are not told, so $f_{v_1}(\mathbf{x}) \neq f_{v_2}(\mathbf{x})$.
- **Four minimums**: three wordings on boundary cases; order randomised between items; the version frozen and dated; one independent open-weight replication.
- **Say why you chose what you chose — one prompt at two temperatures is two instruments.**

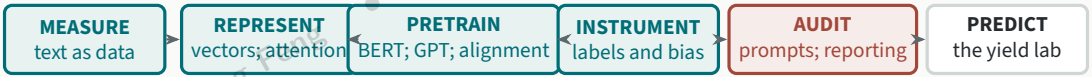
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Source: Chen, Zaharia & Zou, *Harvard Data Science Review* (2024); Miao & Feng, ch. 14, "最低标准".

Session 5.5

Verification, Reproduction, and the Record

What the screening can and cannot see, and what has to be written down



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Three Hallucinations, Three Error Structures

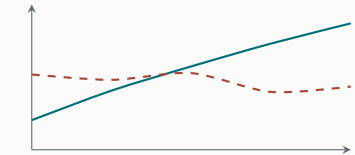
type	what it looks like	error structure	what checks it
factual	the assertion is wrong, the language is not	broad signal noise, varying with the domain	sample, and trace the high-stakes ones
citation	author, year, journal and title all conventional	**variable substitution**: you cite the model's prior	trace each one to the original
numerical	the magnitude is right, the value is wrong	non-classical: correlated with the truth	a range check passes, so go to the raw data

- **A fabricated citation is not noise, and a range check passes on a fabricated number because the model knows the order of magnitude.**

Source: Miao & Feng, *AI for Economists*, ch. 14, table of the three hallucination types.

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Larger Is Not More Truthful — and What That Benchmark Does Not Say

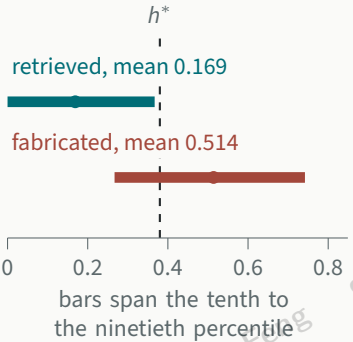


- **Inverse scaling**: as parameters grew, general capability rose while truthfulness on common misconceptions did not. Some larger models did worse.
- **What it does not say**: the benchmark is built on popular misconceptions and does not study citation fabrication. “Bigger models invent more citations” is not that finding.
- **So never read fluency as reliability: a precise detail stated with certainty raises the checking priority.**

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Source: Lin, Hilton & Evans, *ACL* (2022); Miao & Feng, ch. 14, the footnote on inverse scaling.

Self-Consistency Screening

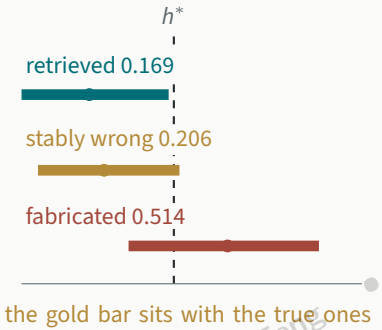


- Sample one prompt R times at non-zero temperature and score the disagreement, $\hat{h} = 1 - \frac{1}{R-1}\sum_{j \geq 2} \mathrm{sim}(s_1, s_j)$.
- Fabricated statements average 0.514, retrieved ones 0.169. Not a clean split: 0.267 sits below 0.367.
- At $h^* = 0.38$, precision 0.745 and recall 0.755: 24.5 per cent of fabrications get through.
- **A larger R steadies the score rather than separating the two, and $R = 8$ is where it stops paying.**

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Source: Manakul, Liusie & Gales, *EMNLP* (2023); the book's own simulation on 600 claims, ch. 14.

What the Screening Structurally Cannot See



- A stably held wrong fact makes every sample agree, so $\hat{h}$ is as low as for a true one: 88.8 per cent of them go unflagged.
- **It measures the variance of fabrication, not truth.** It orders human checking; it is not a test.
- This course's run says it from the other side: the byte-identical prompt agreed with itself 81 per cent of the time, so the variance here is sampling noise.
- **Only a source catches a stable error.**

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claude-sonnet-5 via Claude Code 2.1.278 · 2026-09-20 · prompt 846fe144

Source: Miao & Feng, *AI for Economists*, ch. 14, "自洽性筛查"; Lewis et al. (2020) for the retrieval route.

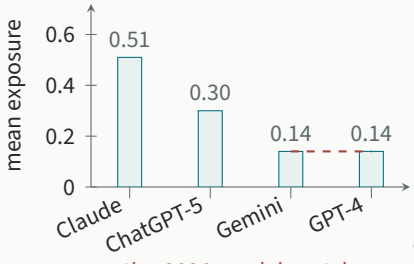
Drift, Hardware, and When Averaging Runs Stops Working

- Two versions of one service are two instruments, $f_{v_1}(\mathbf{x}) \neq f_{v_2}(\mathbf{x})$, under a single name. Record the vendor, the exact version, the system prompt, the sampling parameters and the dates.
- Even with the version fixed and the temperature at zero, bit-exact reproduction is not promised: a parallel reduction reorders the floating-point sums, and addition is not associative.
- A last-digit difference can reorder the softmax and change which token wins, so set the standard where it belongs: the *conclusion* reproduces, not the text.
- Averaging R calls divides the variance by about R — *if* the errors are independent.
- **If the error comes from $\mathcal{P}$, every call makes it, and a larger R computes a biased estimate more precisely.**

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Source: Miao & Feng, *AI for Economists*, ch. 14, "结果复现规范" and the box on hardware non-determinism.

Three Models, One Score Sheet



- Same rubric, same 18,797 tasks, temperature zero; only the rater changes: exposure runs 0.14 to 0.51, a 3.6-fold range, and κ between models 0.36–0.56.
- **The sharp point**: one 2026 model reproduces the original GPT-4 mean almost exactly; the provider's own successor more than doubles it.
- **Downstream, coefficients vary 2.4-fold, and the one significant county estimate is a single 2026 annotator's.**

Source: Yin, Vu & Persico, NBER Working Paper 35110 (April 2026), Tables 1 and 2.

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Three Generations of One Instrument

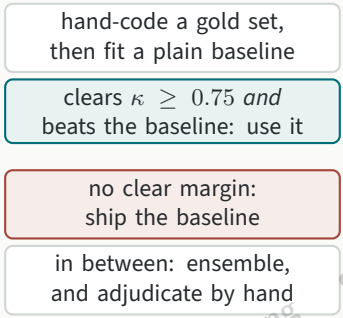
generation	the instrument	re-run next year	what you can audit
ability mapping	a hand-curated map from benchmarks to task abilities	identical	the raters' notes
patent overlap	term-frequency overlap of patents with task text	identical	the patent and the task
model as rater	one model call per task, on the published rubric	**a different number**	the stated reason

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- The first two treat the instrument as a *document*, and documents do not drift.
- **The third treats it as a model call, and a model call has a version.**

Source: Felten, Raj & Seamans (2018, 2021); Webb (2020); Eloundou, Manning, Mishkin & Rock (2023).

When Not to Reach for a Model as Rater



- Hand-code a gold standard, then fit a plain baseline — term frequencies and a logistic regression — and report its κ against that gold.
- Run the model in parallel. If it does not beat the baseline by a real margin, the baseline wins.
- **Name the scheme whenever a κ gets an adjective**: the $\kappa \geq 0.75$ threshold is Fleiss's, Landis and Koch have no such cut, and a third scheme is in circulation.

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Source: Fleiss (1981); Landis & Koch, *Biometrics* 33(1):159–174 (1977); McHugh, *Biochemia Medica* (2012).

What the Policy Requires, What the Book Recommends

journal policy
four obligations, plus one expectation about the reviewer's reading

say in the paper which is which

the book's checklist
nine further items, and the book calls them its own

- The policy's artificial-intelligence section runs to five sentences on a page carrying no date: no AI author; sole accountability with fact-checking; a brief description at submission; disclosure in review, and only tools that will not train on the manuscript.
- It is not a ban on reviewers; the book counts four clauses, leaving out the sentence on the reviewer's own reading.
- **Session 6 carries this into multi-step chains, where the per-step log becomes the record.**

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Source: American Economic Association, *AER* editorial policy, "Use of Artificial Intelligence", read 2026-09-20; Miao & Feng, ch. 14.